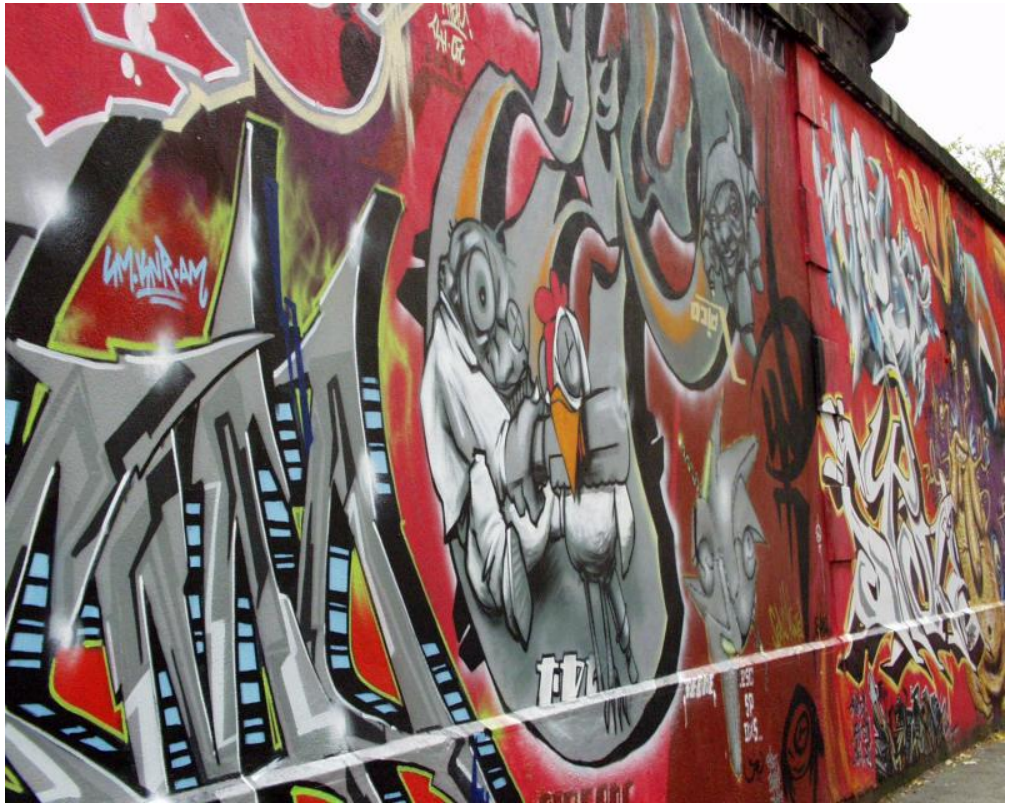
CS5187: Vision & Image

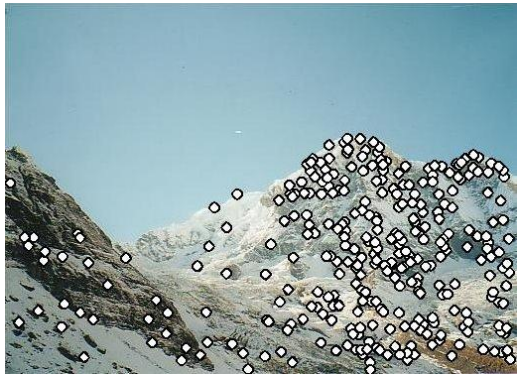
Feature invariance



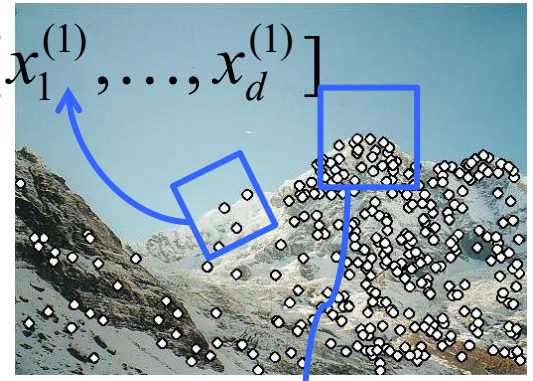


Local features: main components

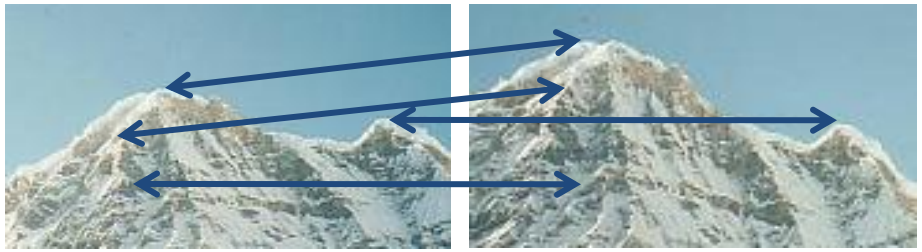
1) **Detection**: Identify the interest points



2) **Description**: Extract vector feature descriptor surrounding each interest point.



3) **Matching**: Determine correspondence between descriptors in two views



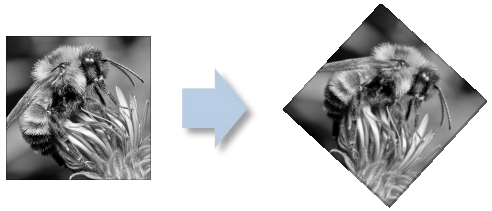
Harris features (in red)



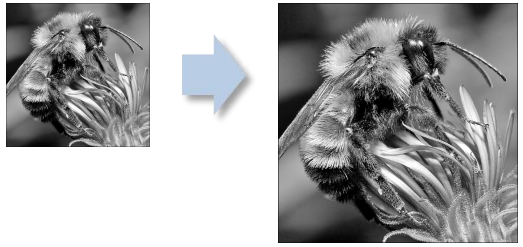
Image transformations

- Geometric

Rotation

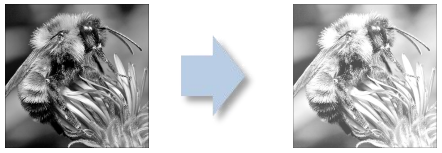


Scale



- Photometric

Intensity change

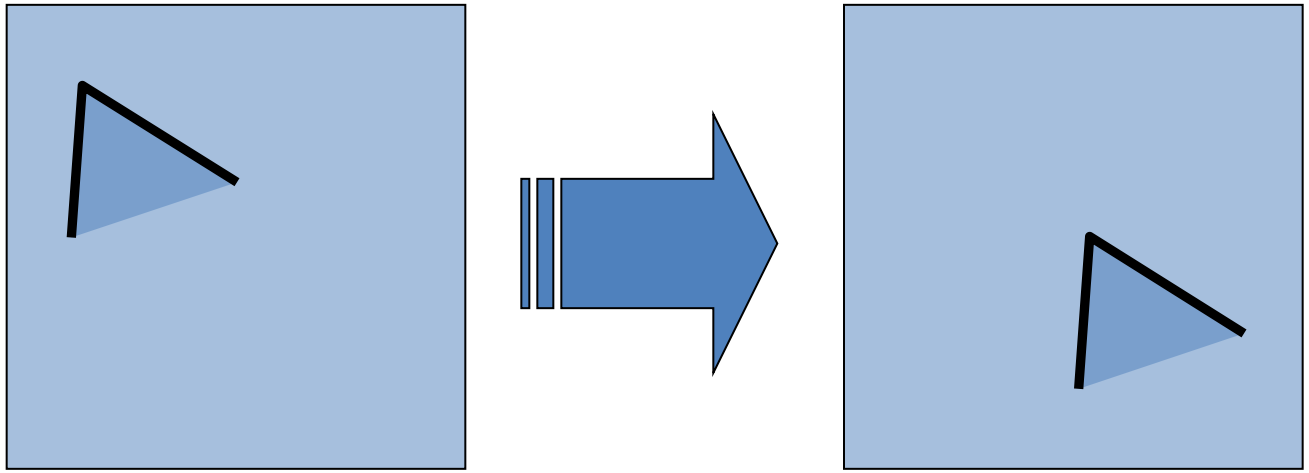


Invariance and equivariance

- We want corner locations to be *invariant* to photometric transformations and *equivariant* to geometric transformations
  - **Invariance:** image is transformed and corner locations do not change
  - **Equivariance:** if we have two transformed versions of the same image, features should be detected in corresponding locations
  - (Sometimes "invariant" and "equivariant" are both referred to as "invariant")
  - (Sometimes "equivariant" is called "covariant")



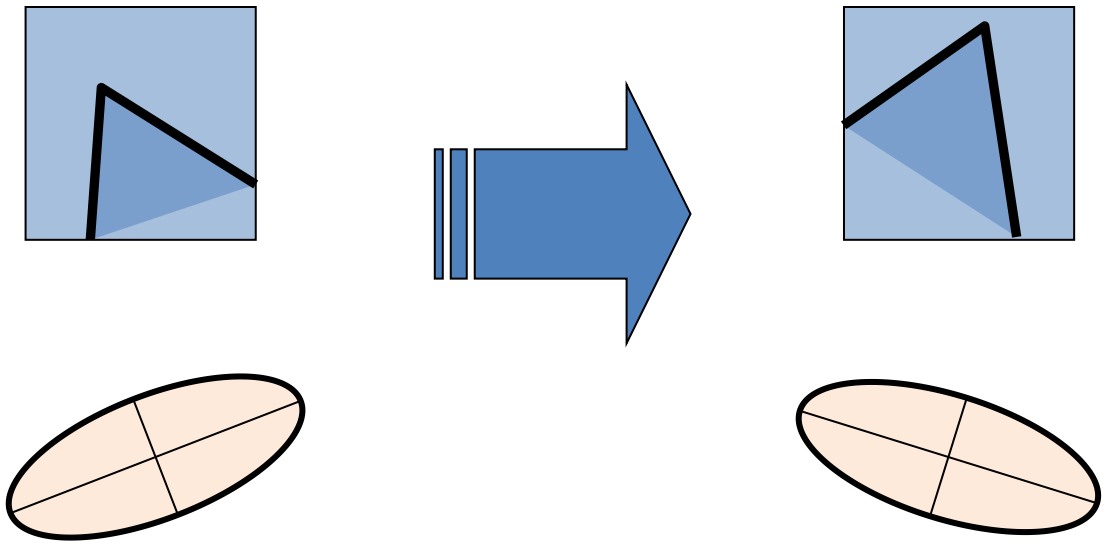
Harris detector invariance properties: image translation



- Derivatives and window function are equivariant

Corner location is equivariant w.r.t. translation

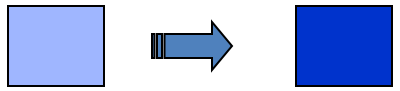
Harris detector invariance properties: image rotation



Second moment ellipse rotates but its shape (i.e. eigenvalues) remains the same

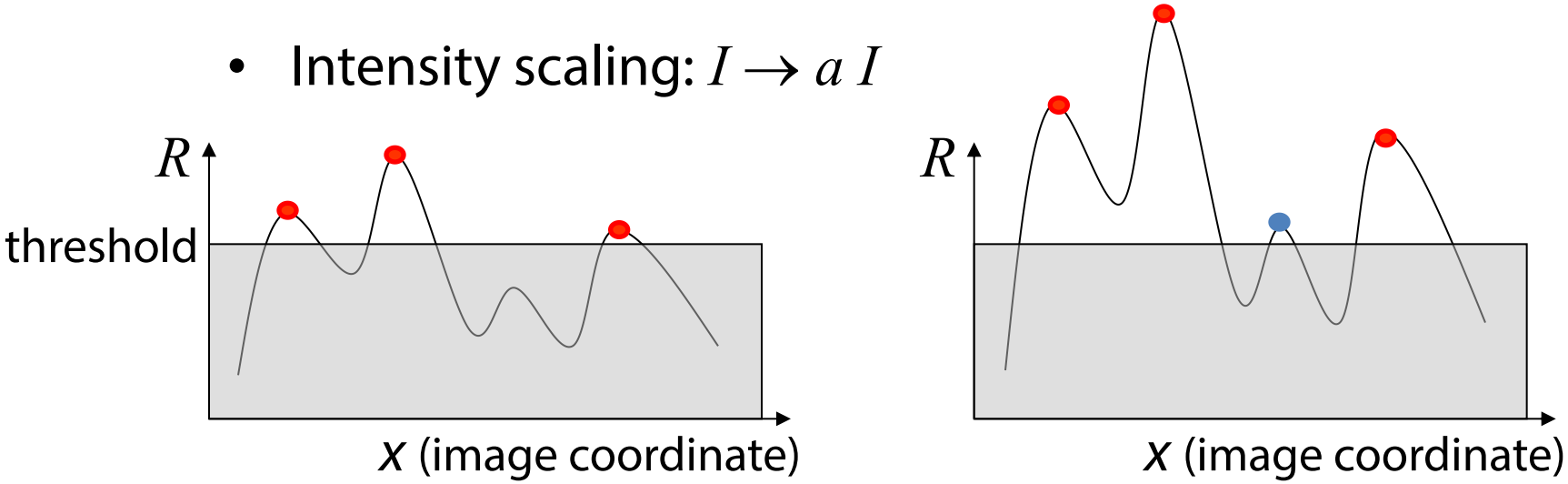
Corner location is equivariant w.r.t. image rotation

Harris detector invariance properties: Affine intensity change



$$I \rightarrow a\,I + b$$

- Only derivatives are used to compute Harris scores → *invariance* to intensity shift $I \rightarrow I + b$
- Intensity scaling: $I \rightarrow a\,I$



Partially invariant to affine intensity change